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Specific heat capacity (RP)

Energy
Paper 1 · Combined and Separate
13 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**State the three variables that a change in thermal energy depends on.**	Mass, specific heat capacity and change in temperature	☐
2	**What does Δθ represent?**	Change in temperature	☐
3 ★	**State the standard unit of specific heat capacity.**	Joule per kilogram per degree Celsius (J/kg °C)	☐
4 ★	**Define specific heat capacity.**	The energy required to raise the temperature of 1 kg of a substance by 1 °C.	☐
5	**A material has a high specific heat capacity. Is it hard or easy to heat up, and why?**	Hard. It takes a lot of energy to raise the temperature of 1 kg by 1 °C.	☐
6 ★	**State the three measurements needed to find the specific heat capacity of a substance.**	Energy supplied, mass and temperature rise	☐
7	**Name the piece of equipment used to measure temperature.**	A thermometer	☐
8	**Name the piece of equipment used to measure mass.**	A balance	☐
9	**Explain why a Bunsen burner is not used when measuring specific heat capacity.**	The energy it supplies cannot be measured easily.	☐
10 ★	**Explain why an electrical heater is used when measuring specific heat capacity.**	The energy supplied can be calculated from potential difference, current and time.	☐
11	**State the equation linking change in thermal energy, mass, specific heat capacity and temperature change.**	change in thermal energy = mass × specific heat capacity × temperature change ($\Delta E = m\ c\ \Delta\theta$)	☐
12 ★	**When measuring specific heat capacity, explain why the block is wrapped in insulation.**	To reduce energy transfer to the surroundings.	☐

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	QUESTION	ANSWER	✓
13 ★	**Explain why a measured specific heat capacity is usually higher than the true value.**	Some of the energy supplied is transferred to the surroundings, not the block.	☐

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